

SmartLearners.ai — Knowledge Transfer

Overview

The system is a single FastAPI backend (`main.py`) that powers **three distinct chatbots** sharing the same infrastructure. All three use Google Gemini AI under the hood and follow the same session-based pattern.

Chatbot 1 — Student Chatbot

Who uses it: Students (Class 6–12), via a mobile/web app.

Session type: `user_type = "student"`

What it can do:

Feature	What it does
Performance Analysis	Tells the student how they did overall — scores, grades, percentages — across exams, homework, and self-practice.
Strength Detection	Identifies topics and questions where the student performed well (100% score, no errors).
Weakness Detection	Identifies where the student struggled — broken down by error type (conceptual mistake, calculation mistake, skipped, incomplete).
Specific Exam Drill-down	Student can ask about a specific exam (e.g., "SA1") and get results only for that exam.
Question-wise Breakdown	Shows performance on each individual question — what went wrong and what concepts were involved.
Remedial Study Plan	Generates a day-by-day study schedule for weak chapters, prioritized by how important the chapter is in NCERT exams and how often the student got it wrong.
Tutor Mode — Solve a Problem	The student sends a question and the bot solves it step by step, like a tutor.
Tutor Mode — Check My Answer	The student submits their own answer and the bot tells them if it's right or wrong and why.
Image Support	Student can upload a photo of a question (e.g., a handwritten problem) and ask about it.
Voice Input	Student can send an audio message — it gets transcribed and processed as a text query.
Multi-language Support	Detects and responds in English, Hindi, or Telugu.
Guardrails	Blocks off-topic questions (non-academic) and attempts to extract system instructions.

Chatbot 2 — Teacher Chatbot

Who uses it: Teachers, via the same app (different login role).

Session type: `user_type = "teacher"`

What it can do:

Feature	What it does
Class Performance Summary	Overview of how all students in a class did on an exam — average, highest, lowest score.
Top Performers	Lists the best-scoring students in the class for any given exam.
Struggling Students	Lists students who scored below a threshold (default 50%) and what they need help with.
Grade Distribution	Shows what percentage of the class got A+, A, B+, B, C, D, F — and the class pass rate.
Common Weak Areas	Finds the topics most students struggled with, so the teacher knows what to re-teach.
Specific Student Drill-down	Teacher can ask about one student by name and get their detailed question-by-question performance.
Class Comparison	Compare performance across different class sections (e.g., 10A vs. 10B).
Teaching Recommendations	Suggests what the teacher should focus on in the next class based on class-wide weaknesses.
Guardrails	Blocks off-topic questions and prompt extraction attempts, same as student bot.

Chatbot 3 — WhatsApp Chatbot

Who uses it: Students (and potentially parents) on WhatsApp via Interakt.

Session type: Stateless — no session ID, identified by phone number.

What it can do:

Feature	What it does
Auto Student Lookup	When a message arrives, the system looks up the student by their phone number in the database. No login required.
Registration Flow	If the phone number is not registered, sends a link to sign up for a free trial.
Performance Queries	Same analysis as the Student Chatbot — scores, weaknesses, strengths — but delivered over WhatsApp.
Simple Responses	Greetings, "who are you?", "help" — answered instantly without calling the AI.

Multi-language	Detects the language of the message and replies in the same language (English/Hindi/Telugu).
WhatsApp Formatting	Responses are formatted with WhatsApp bold (<i>*text*</i>) instead of markdown, so they display correctly in the app.
Async Processing	Returns a 200 OK to Interakt immediately, then processes and replies in the background — so WhatsApp doesn't time out.

Bonus — Test Prep Analysis

Not a chatbot, but a dedicated endpoint: `/test-prep-analysis`

Takes a student's MCQ exam results (with right/wrong answers), finds all wrong questions, and for each one:

- Explains the concept behind the question
- Shows exactly where the student's reasoning went wrong
- Generates 2 new practice MCQs on the same concept

This bypasses the full pipeline entirely and goes straight to Gemini with a structured output schema.